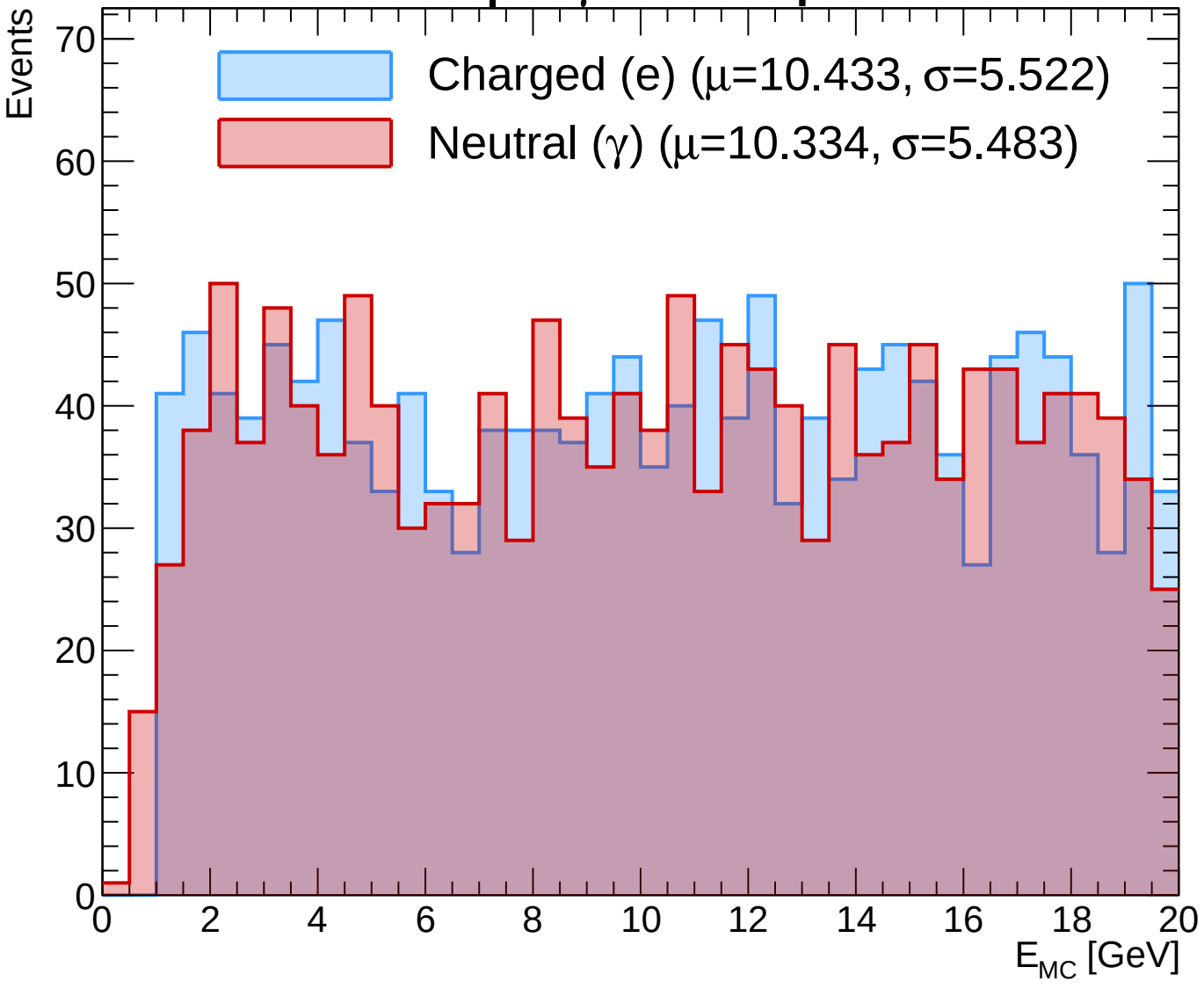
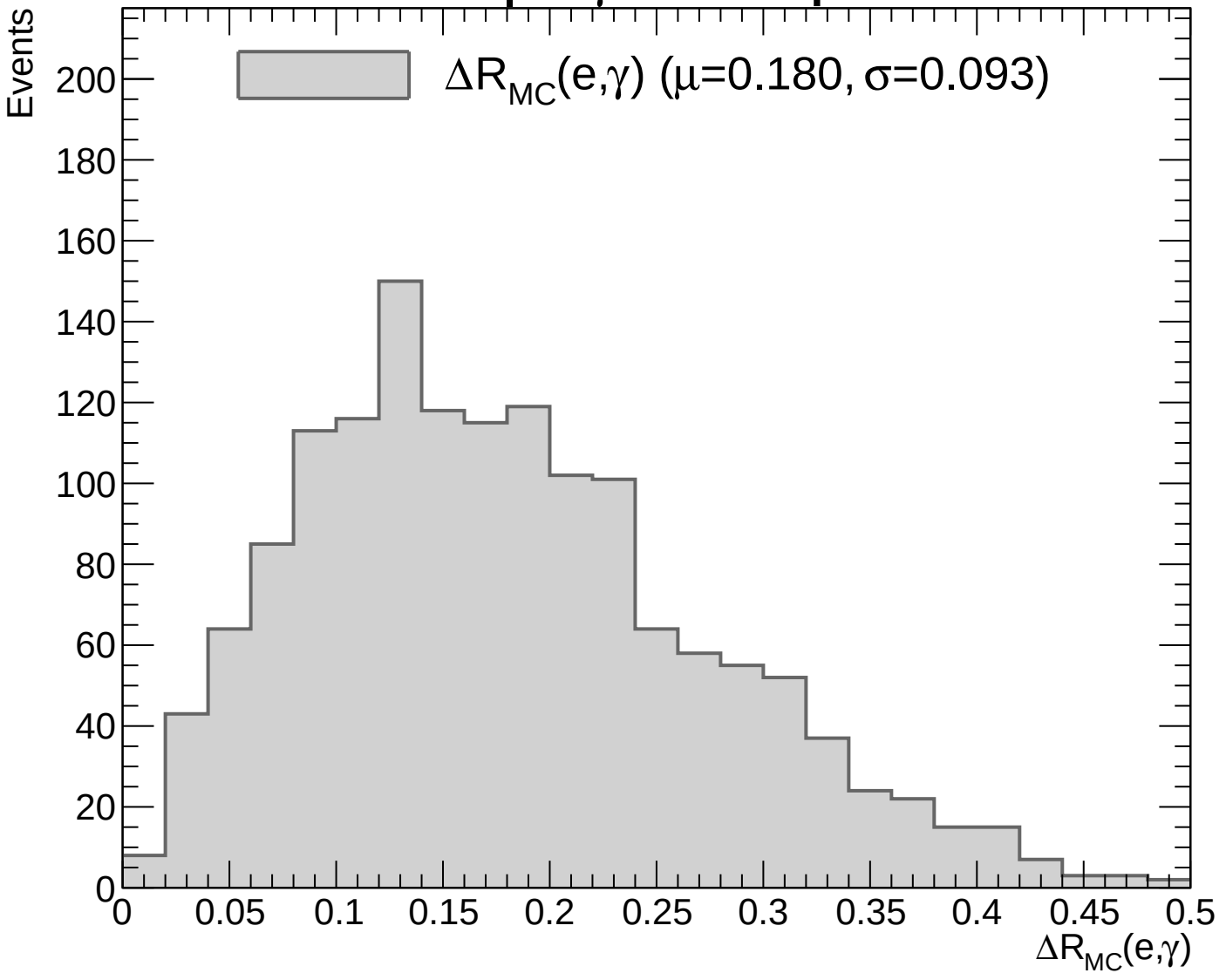
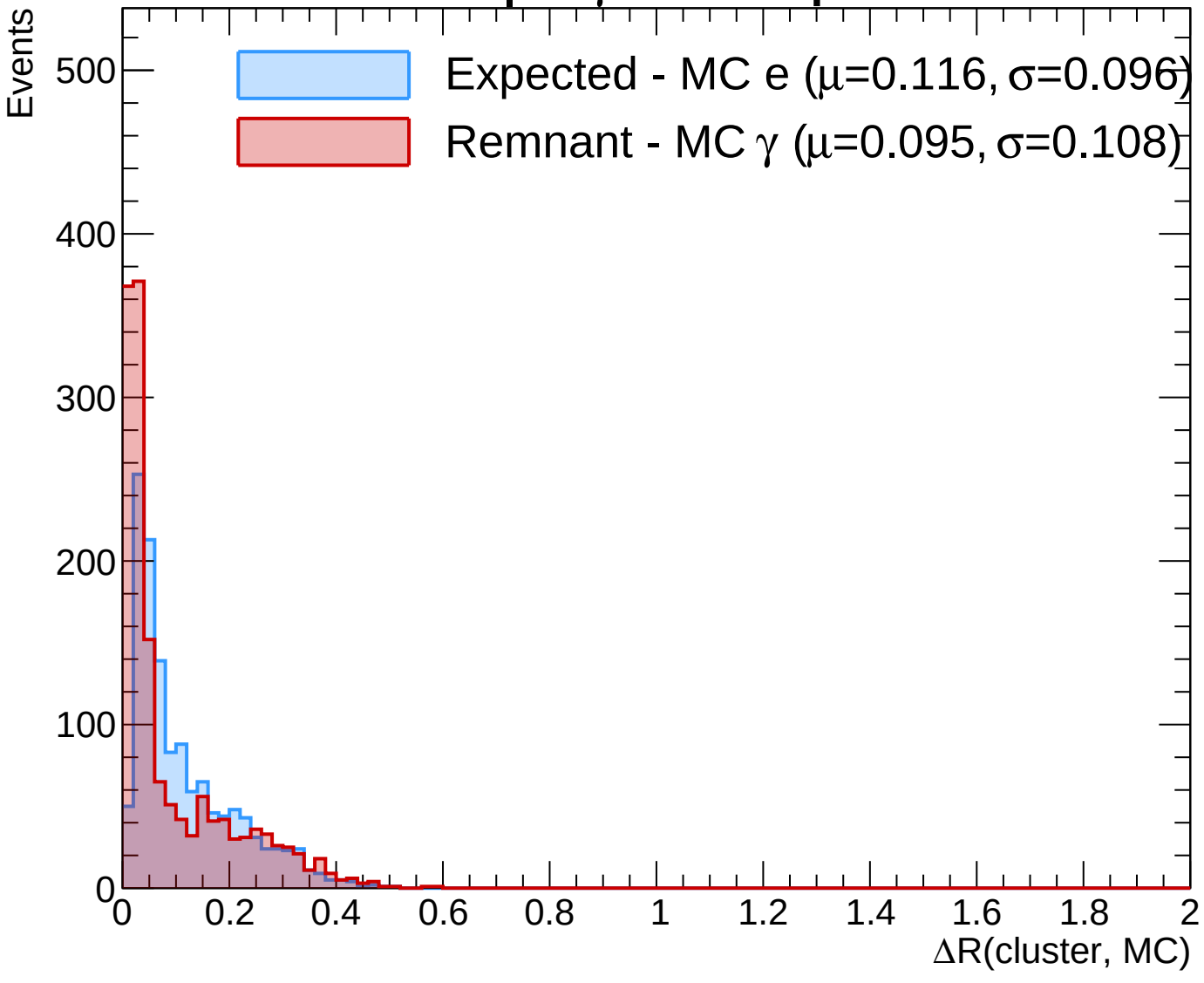
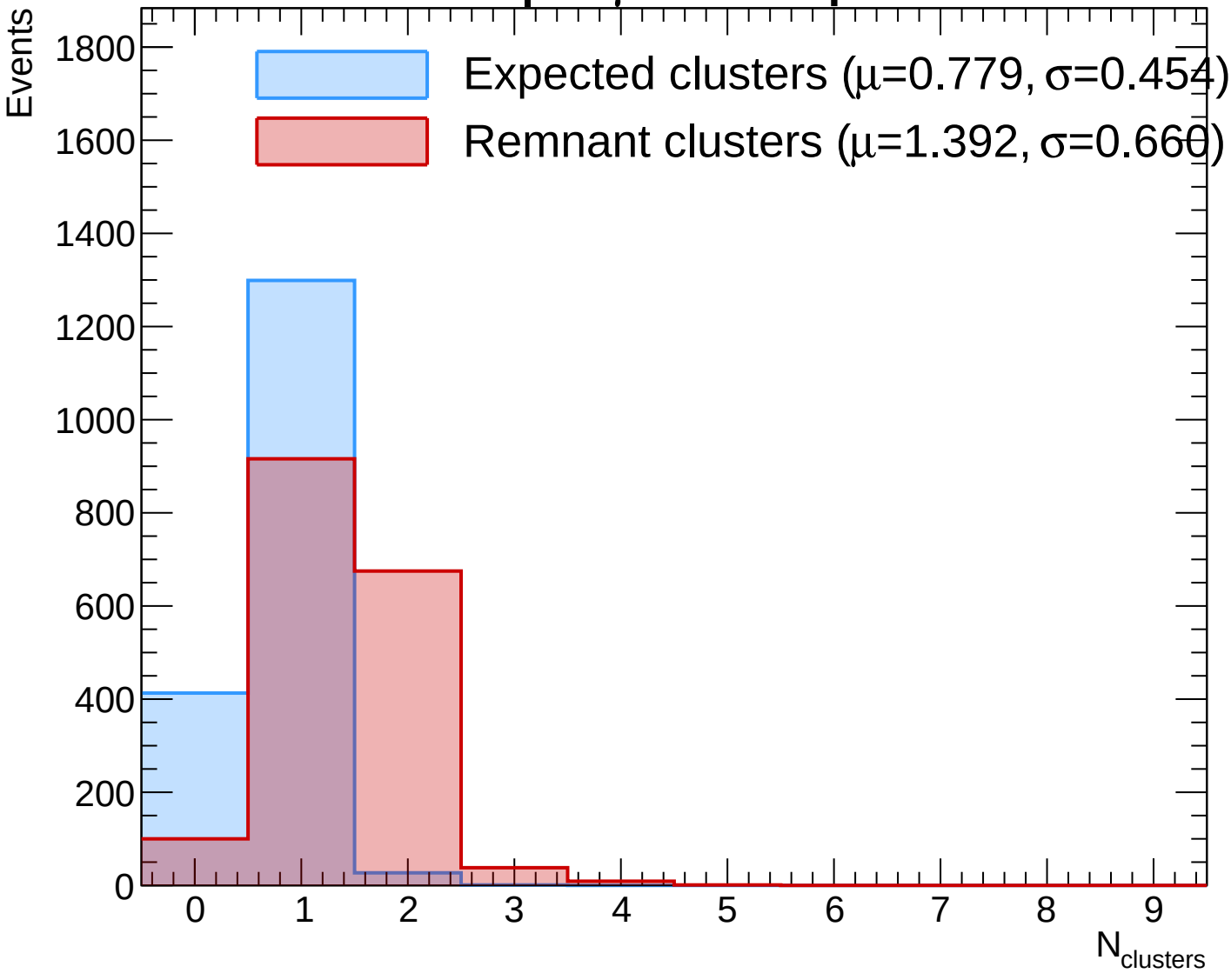
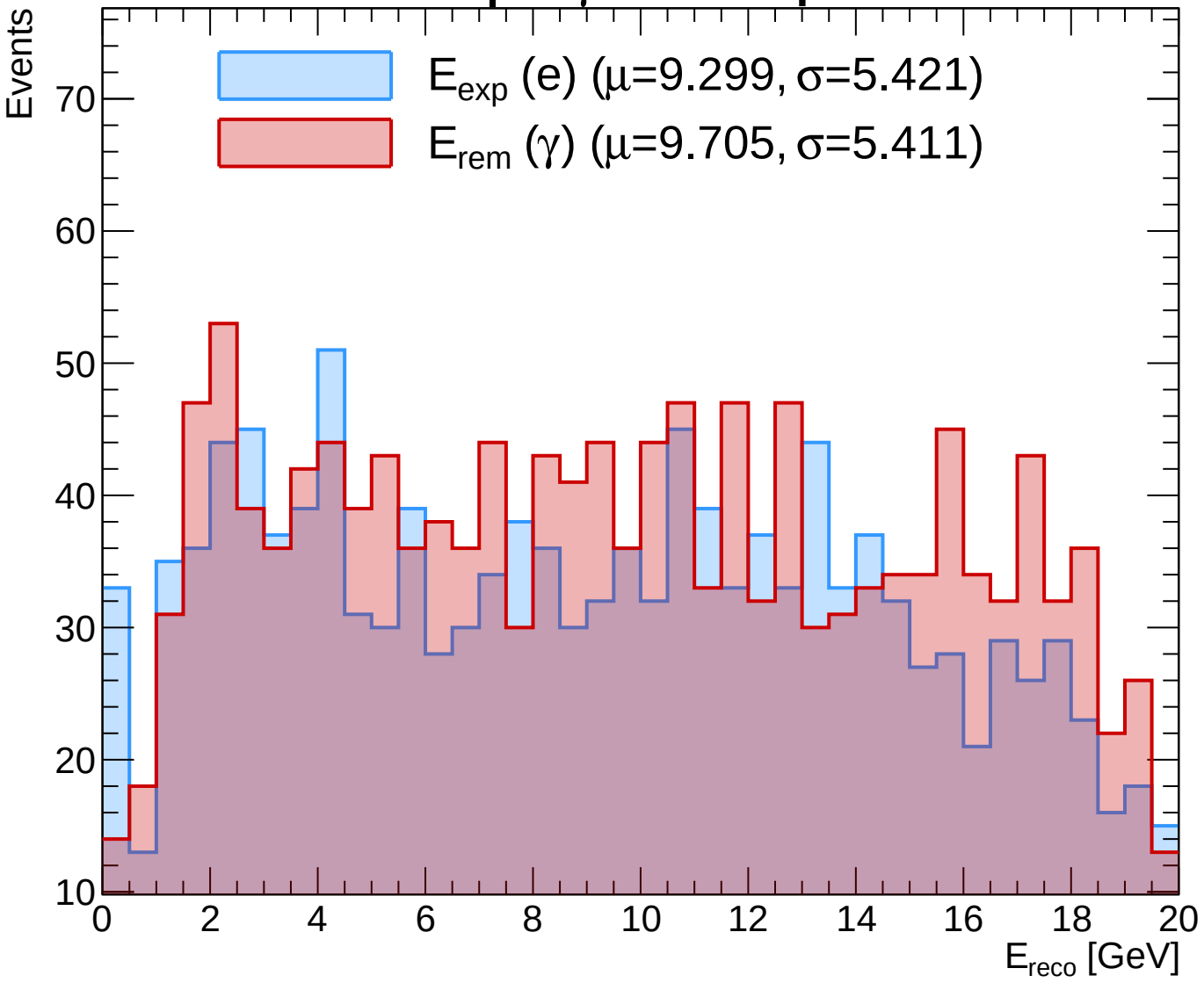
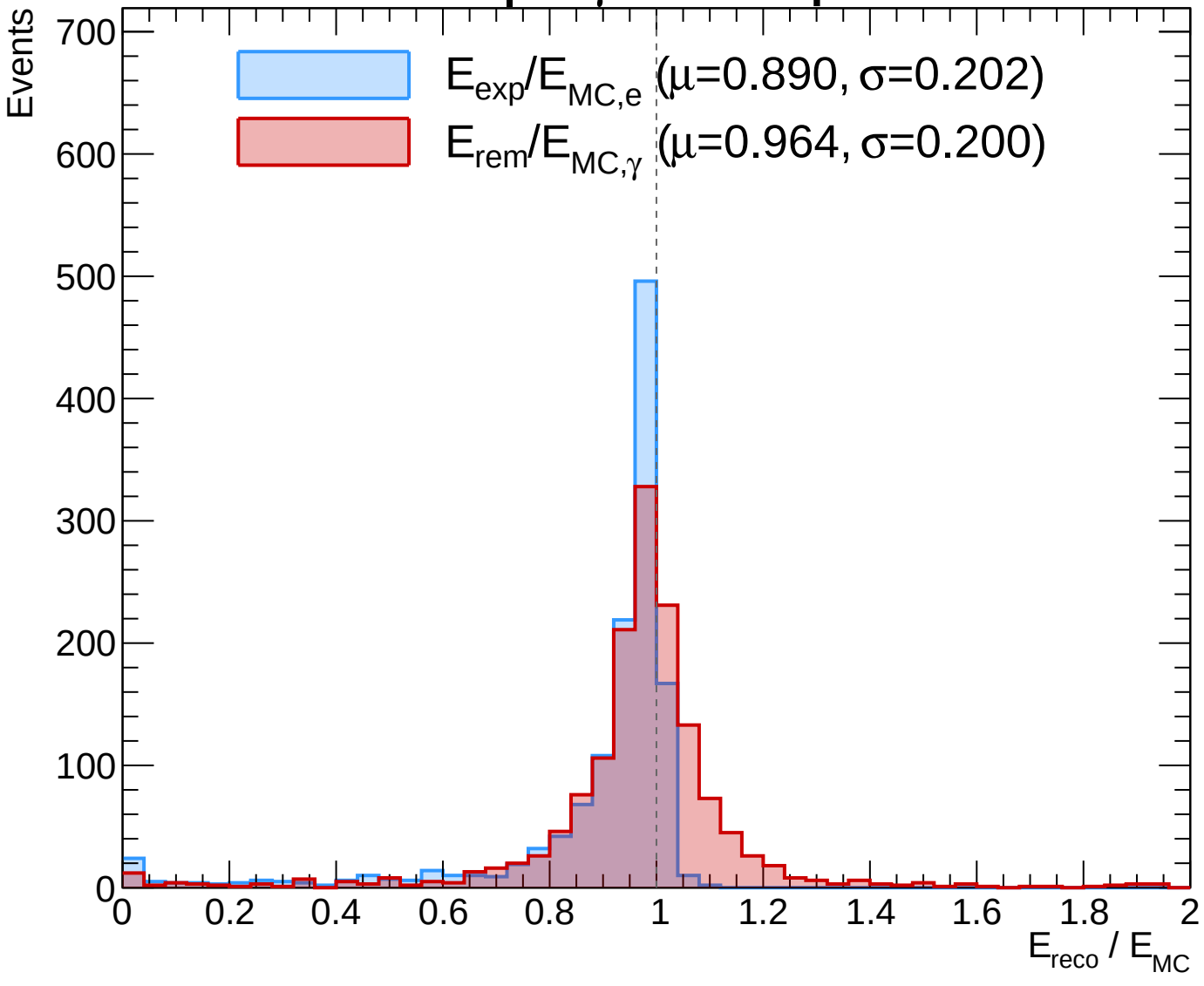

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Charged (e) (μ=10.433, σ=5.522)
Neutral (γ) (μ=10.334, σ=5.483)
0
10
20
30
40
50
60
70
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]

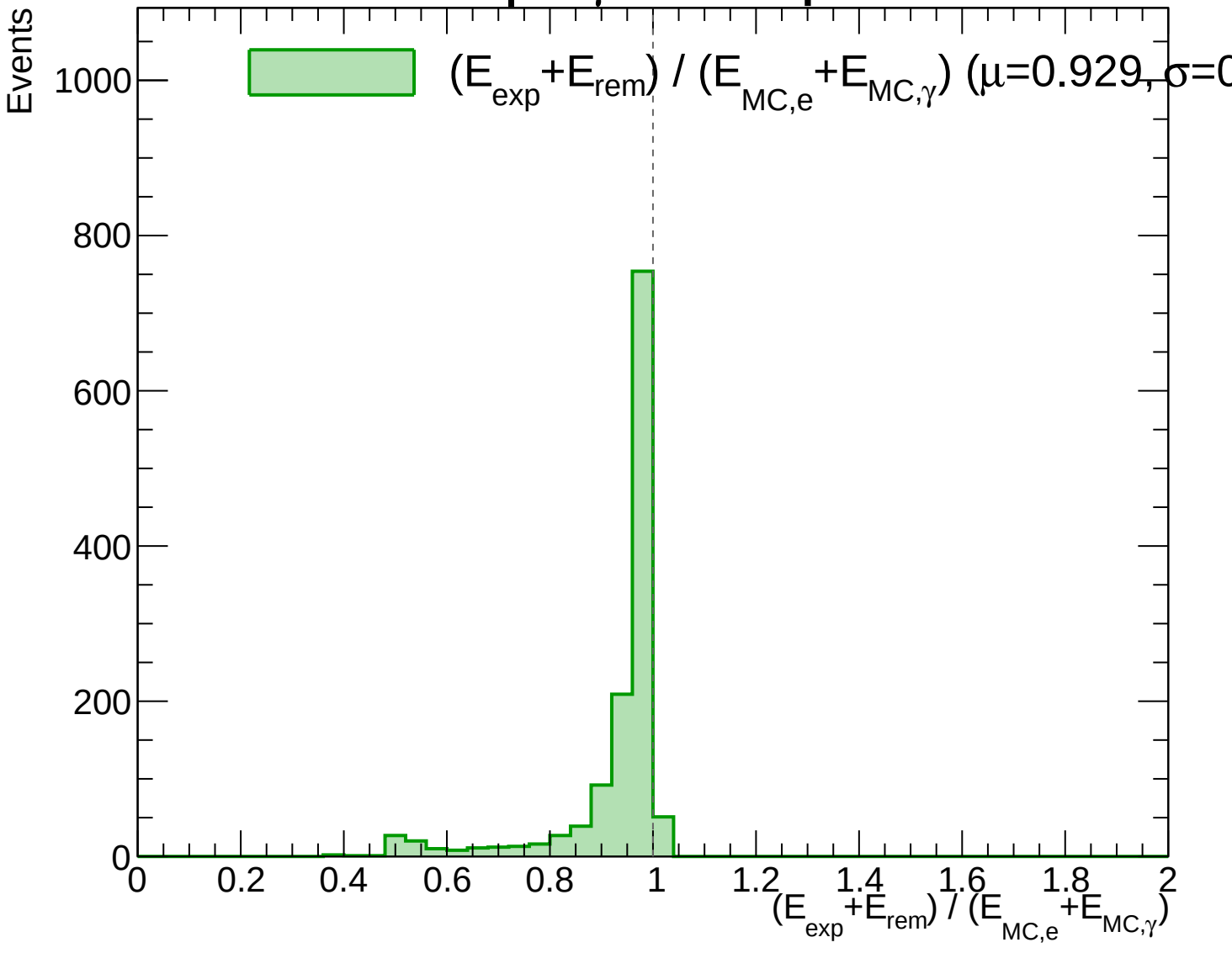

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
ΔR_MC(e,γ) (μ=0.180, σ=0.093)
0
20
40
60
80
100
120
140
160
180
200
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)

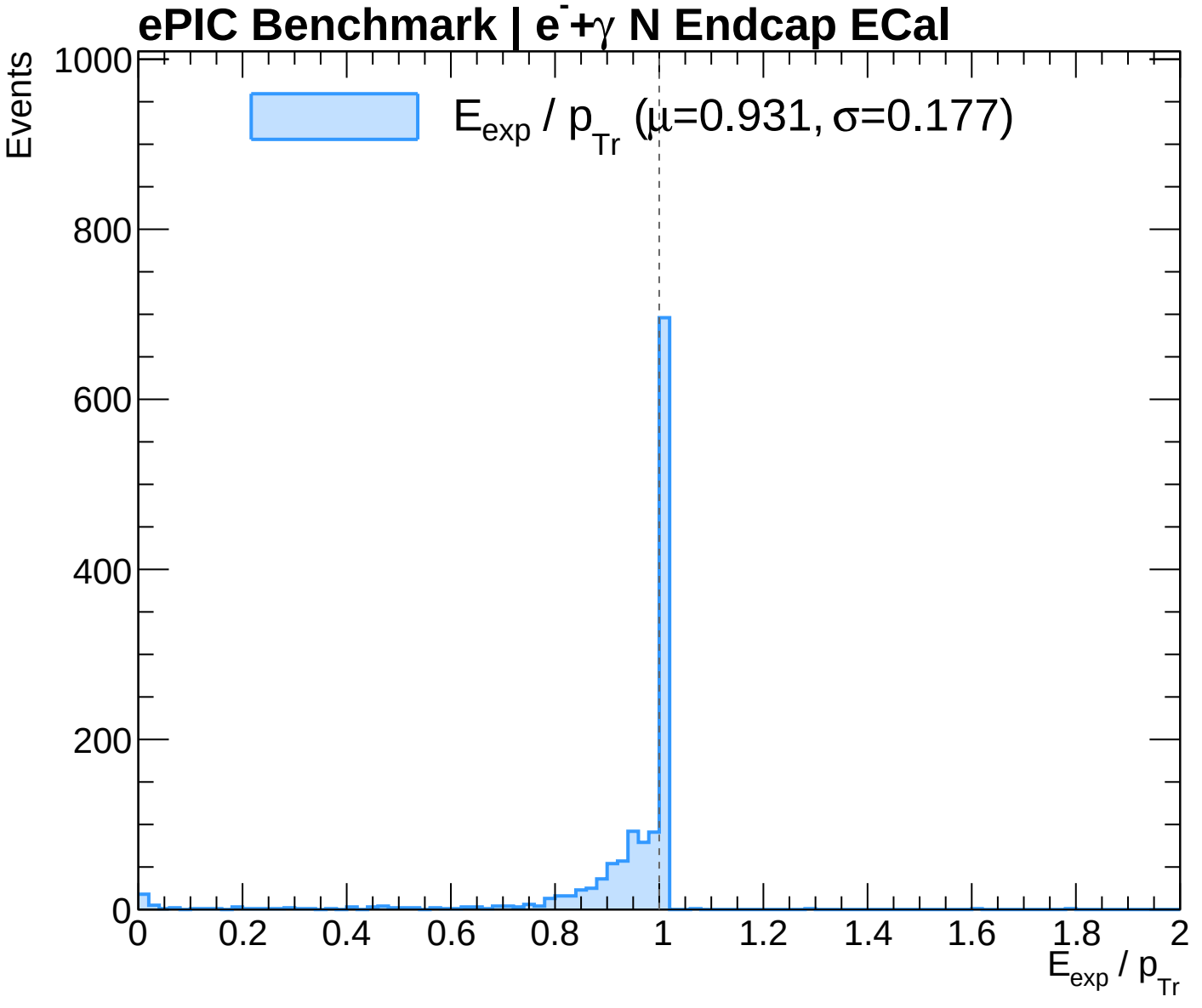

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
Expected - MC e (μ=0.116, σ=0.096)
Remnant - MC γ (μ=0.095, σ=0.108)
0
100
200
300
400
500
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΔR(cluster, MC)

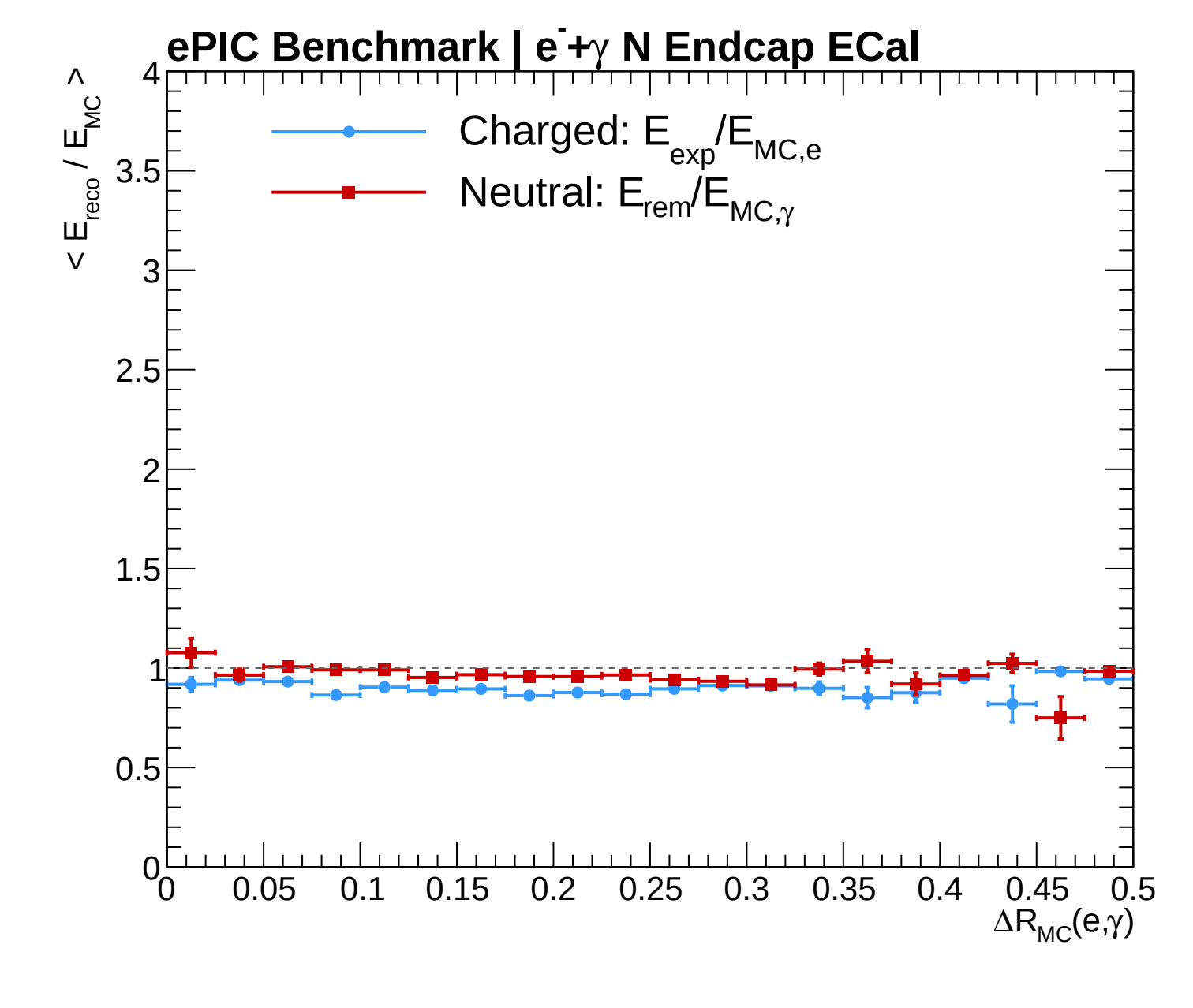

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
1800
1600
1400
1200
1000
800
600
400
200
0
Expected clusters (μ=0.779, σ=0.454)
Remnant clusters (μ=1.392, σ=0.660)
0
1
2
3
4
5
6
7
8
9
N clusters

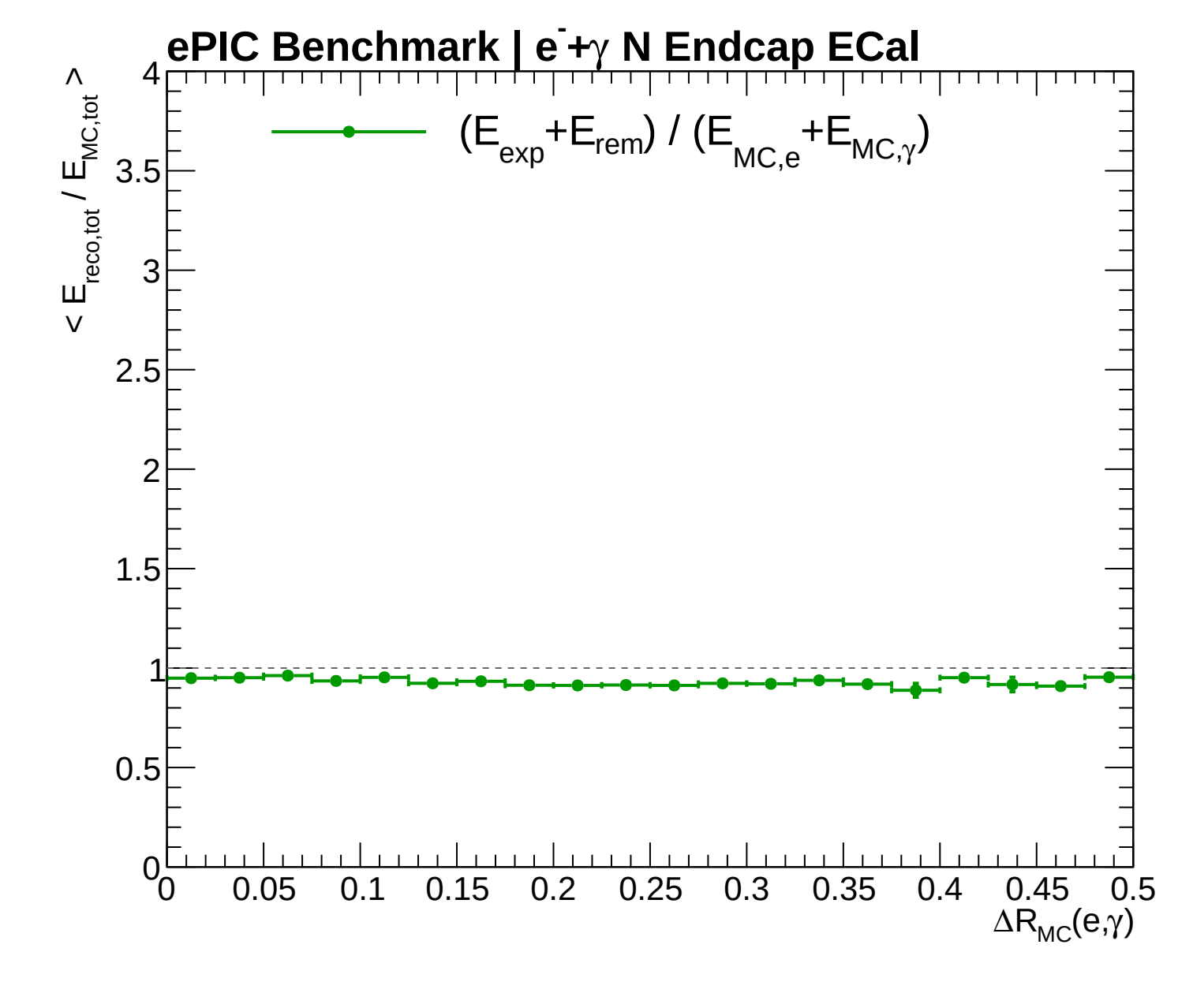

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
E_exp (e) (μ=9.299, σ=5.421)
E_rem (γ) (μ=9.705, σ=5.411)
70
60
50
40
30
20
10
0
2
4
6
8
10
12
14
16
18
20
E_reco [GeV]

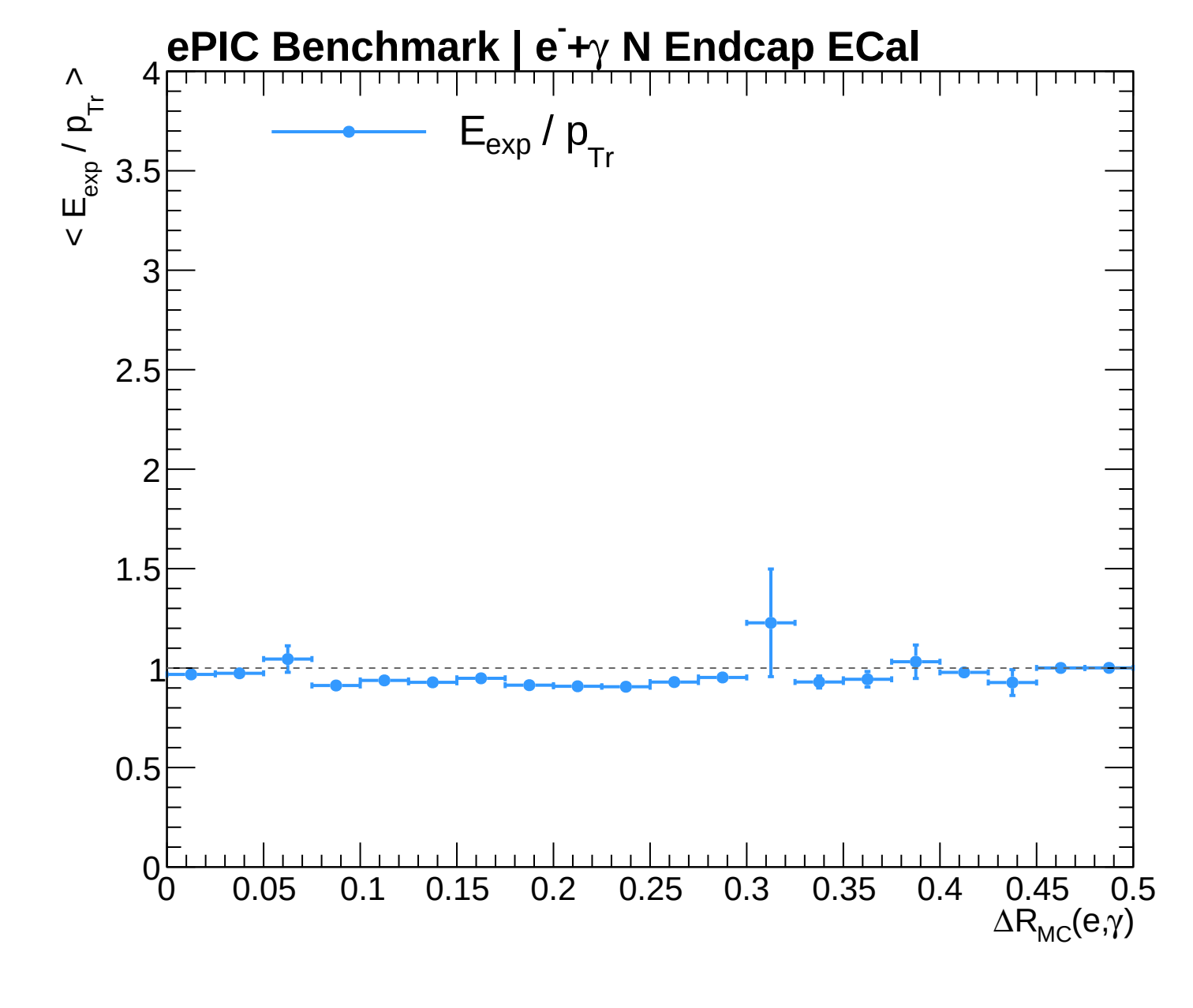

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
700
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_reco / E_MC
E_exp/E_MC,e (μ=0.890, σ=0.202)
E_rem/E_MC,γ (μ=0.964, σ=0.200)

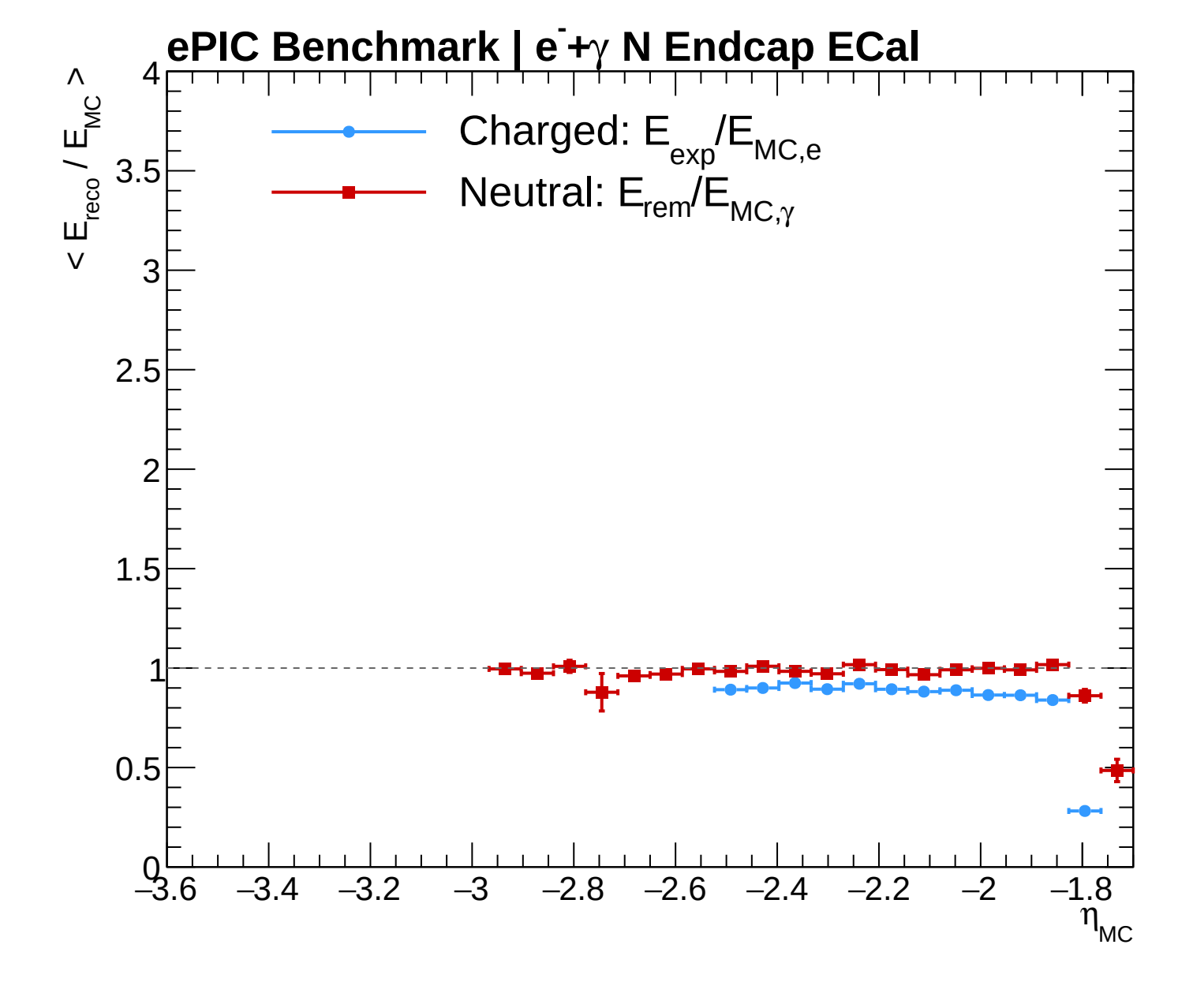

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
(E_exp+E_rem) / (E_MC,e+E_MC,γ) (μ=0.929, σ=0
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_exp+E_rem) / (E_MC,e+E_MC,γ)

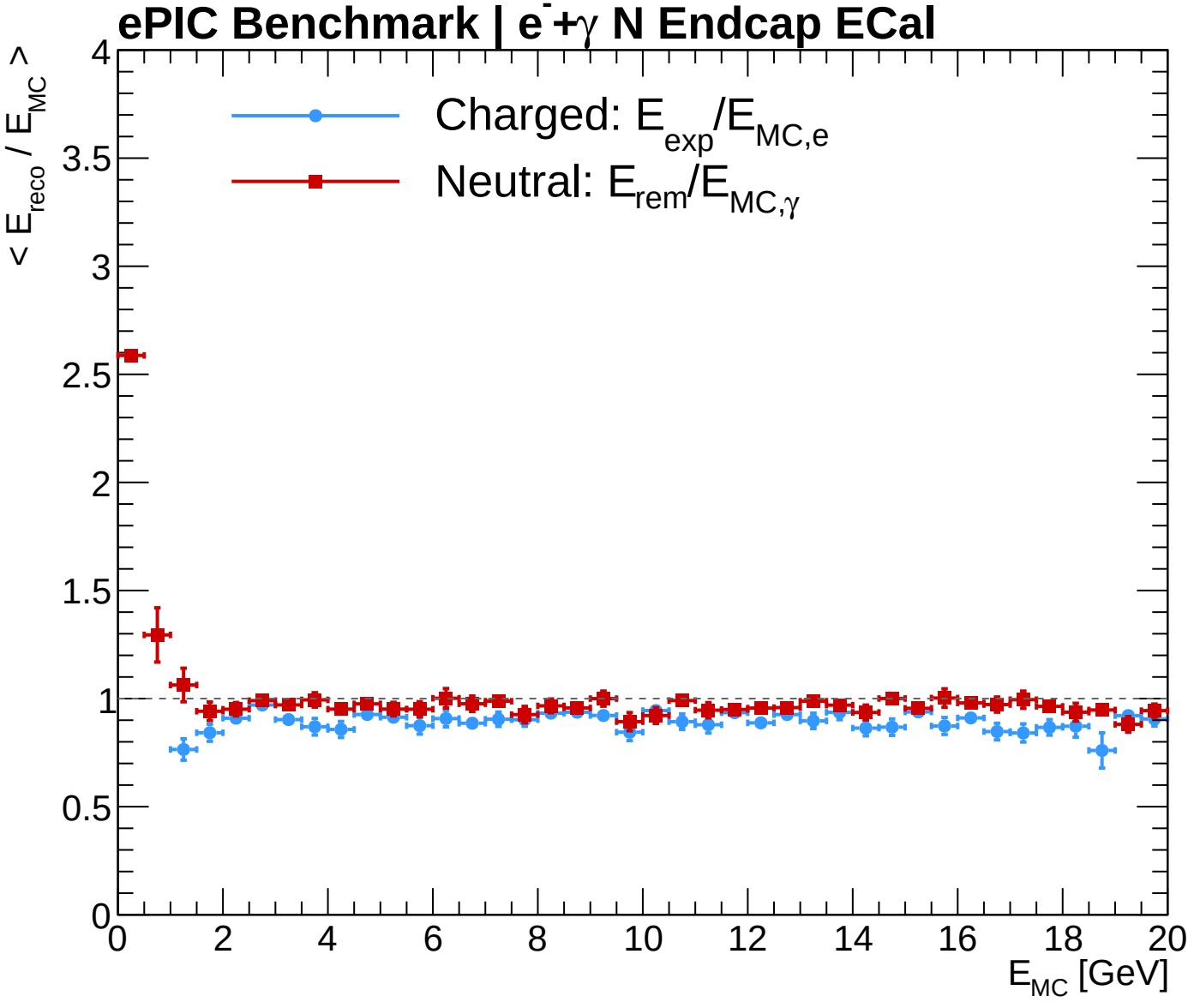

ePIC Benchmark | e⁻+γ N Endcap ECal
Events
1000
800
600
400
200
0
E_exp / p_Tr (μ=0.931, σ=0.177)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

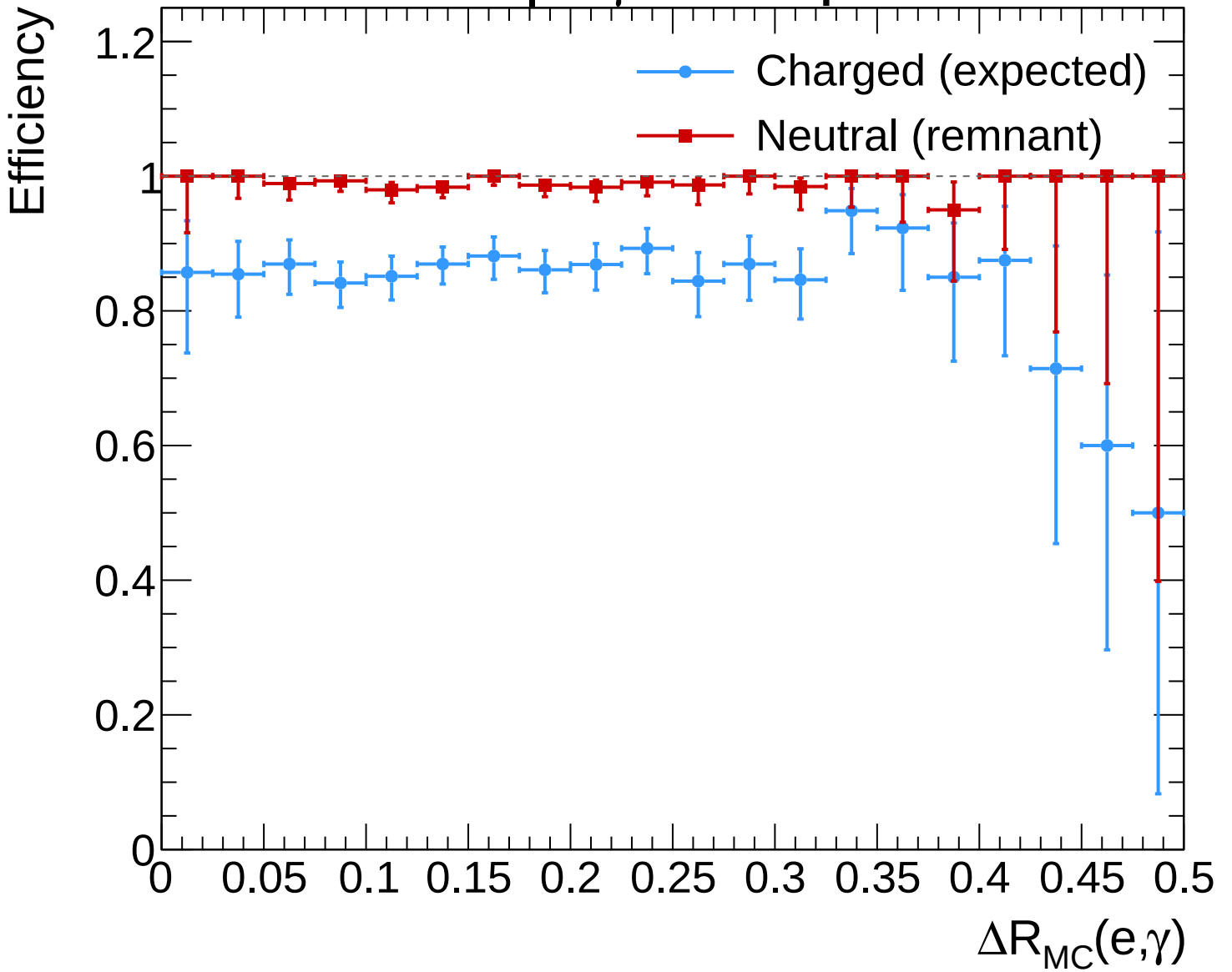

ePIC Benchmark | e⁻+γ N Endcap ECal
Charged: E_exp/E_MC,e
Neutral: E_rem/E_MC,γ
< E_reco / E_MC >
ΔR_MC(e,γ)
0
0.5
1
1.5
2
2.5
3
3.5
4
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

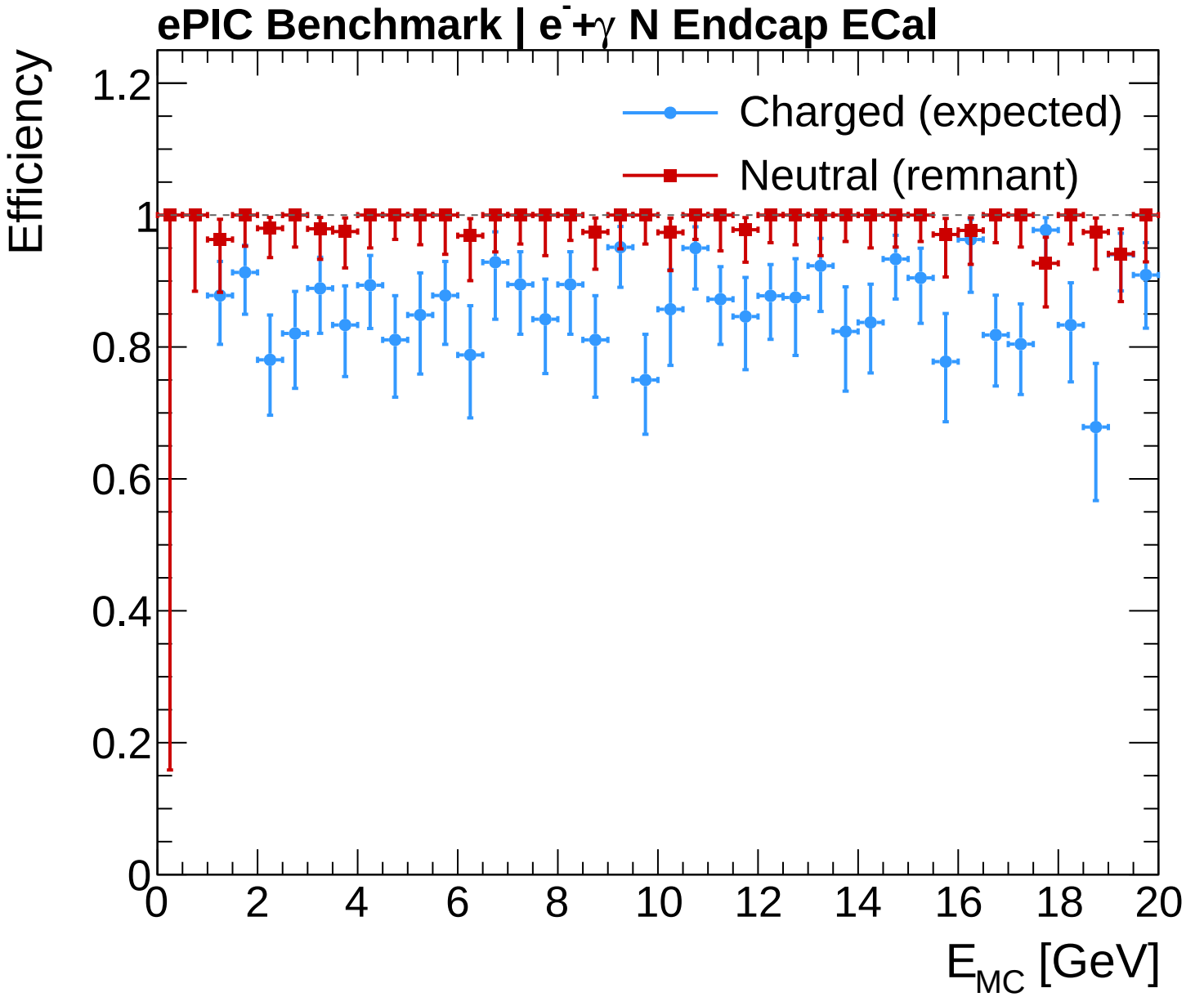

ePIC Benchmark | e⁻+γ N Endcap ECal
(E_exp+E_rem) / (E_MC,e+E_MC,γ)
< E_reco,tot / E_MC,tot >
ΔR_MC(e,γ)
0
0.5
1
1.5
2
2.5
3
3.5
4
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

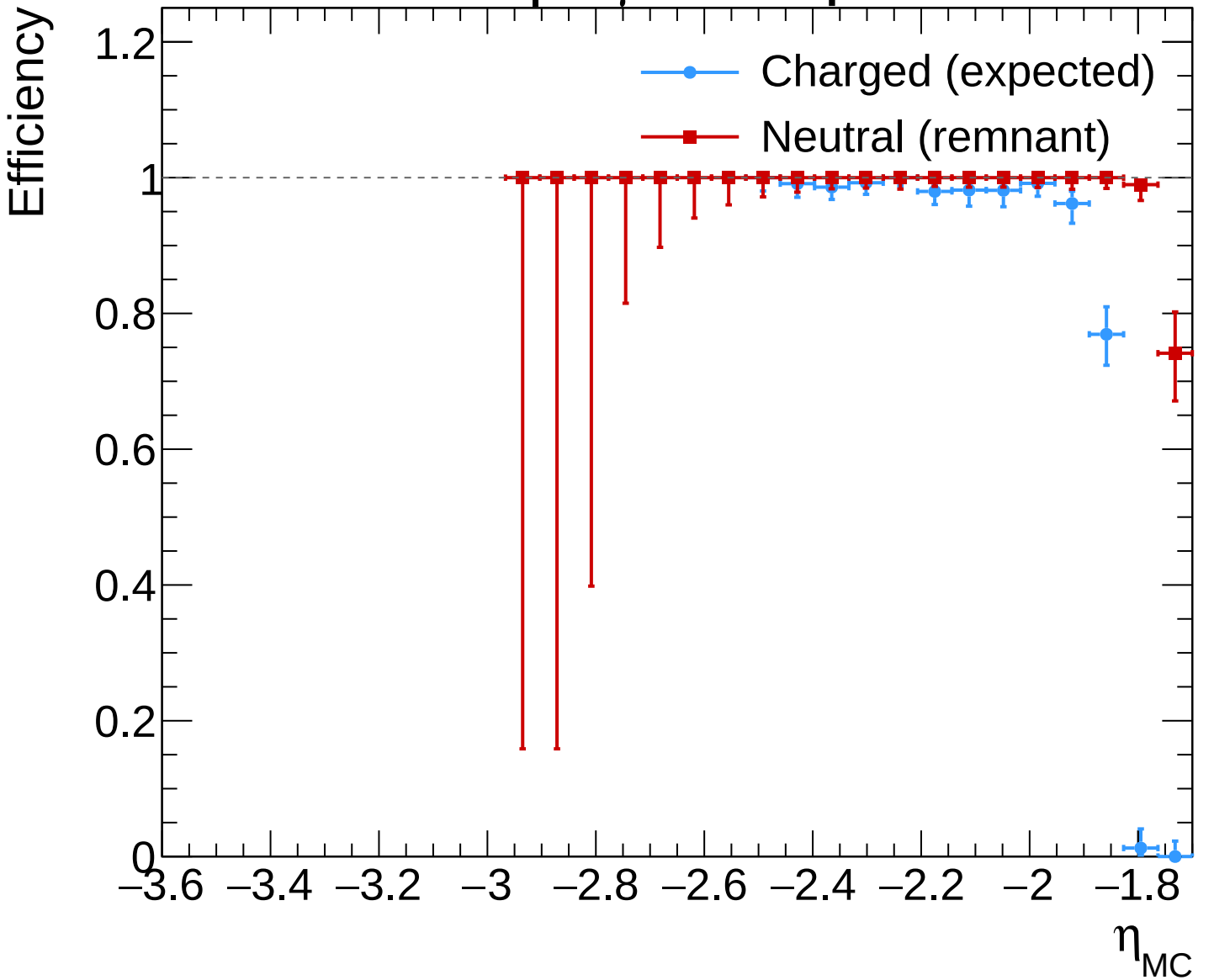

ePIC Benchmark | e⁻+γ N Endcap ECal
< E_exp / p_Tr >
E_exp / p_Tr
4
3.5
3
2.5
2
1.5
1
0.5
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ N Endcap ECal
Charged: E_exp/E_MC,e
Neutral: E_rem/E_MC,γ
< E_reco / E_MC >
η_MC
4
3.5
3
2.5
2
1.5
1
0.5
0
–3.6
–3.4
–3.2
–3
–2.8
–2.6
–2.4
–2.2
–2
–1.8


ePIC Benchmark | e⁻+γ N Endcap ECal
Charged: E_exp/E_MC,e
Neutral: E_rem/E_MC,γ
< E_reco / E_MC >
E_MC [GeV]
0
0.5
1
1.5
2
2.5
3
3.5
4
0
2
4
6
8
10
12
14
16
18
20


ePIC Benchmark | e⁻+γ N Endcap ECal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ N Endcap ECal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]


ePIC Benchmark | e⁻+γ N Endcap ECal
Efficiency
1.2
1
0.8
0.6
0.4
0.2
0
Charged (expected)
Neutral (remnant)
−3.6
−3.4
−3.2
−3
−2.8
−2.6
−2.4
−2.2
−2
−1.8
η_MC